A Large Language Model (LLM) Approach to COVID-19 Fake News Detection

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***Abstract*—Text classification is a widely researched area within the domains of information retrieval and data mining, offering valuable applications in various fields, such as healthcare diagnosis, targeted marketing, entertainment, and the classification of COVID-19-related fake news. This problem has gathered significant attention from researchers globally, particularly due to recent advancements in data mining and natural language processing. These advancements have prompted the development of automated systems to efficiently classify textual data. Natural Language Processing (NLP) enables the classification of documents containing diverse textual content. Social media platforms generate a vast volume of data through user interactions. The methodology of the current research involves gathering a dataset. Within this realm, datasets such as the COVID-19 English Tweet Dataset have been instrumental for experimental purposes. This dataset encompasses various forms of textual content, including tweets, posts, and messages pertaining to the coronavirus pandemic, and also includes speech. In an effort to harness the potential of transfer learning (TL) and large language models (LLMs), the proposed research endeavors to apply pretrained transformer based models and LLMs, namely BERT-base, ALBERT-base, XLM-RoBERTa-base, DistilBERT, GPT-2, XLNet, and T5, to this dataset. Additionally, six different optimization techniques, namely weighted fusion, particle swarm optimization (PSO) fusion, the Nelder–Mead algorithm, Powell's method, the Broyden Fletcher–Goldfarb–Shanno (BFGS) algorithm, and the trust region constrained optimization algorithm, are applied to enhance the models' results. The results section indicates that the GPT-2 large language model demonstrates high accuracy and efficiency. The model's performance is evaluated using metrics such as accuracy, precision, recall, and F1-score, with a score of 0.96 achieved for each metric. Moreover, heatmaps are created to visualize the conclusions obtained from each model. Finally, the conclusion of the research study highlights the value of applying LLMs and identifies the optimal performer among all the classifiers. Lastly, the study outlines future directions for further exploration and enhancement in this domain.**

***Index Terms*—COVID-19, fake news detection, large language models, transfer learning, transformers, optimization**

I. Introduction

Natural Language Processing (NLP) is defined as a scientific discipline, which aims at providing computers with the ability to understand and generate natural language. NLP has recently received further attention given the expansion of the studies in the area and the increased computational capabilities. There are now efforts to find out meaningful data hidden in language as a useful tool in education, healthcare, industries, and retail. First, it was employing rule-based methods to solve NLP problems, which gave way to machine learning (ML) methods due to the richness of human language. Therefore, these methods are more effective in addressing the notions and dynamics of natural language. These methods include Support Vector Machine (SVM) and Naïve Bayes classifiers.

Text mining is connected with Natural Language Processing (NLP) concept which deals with getting valuable information from the text present in social media platforms. This type of extraction is achieved through algorithms and methods that work with textual data. Despite being an essential part of AI, NLP relies on advanced approaches such as text mining and data analysis. Data mining, an integration of machine learning and deep learning, is a versatile tool with high success rates in several areas. It improves question answering, anaphora resolution, bioinformatics, and web based network analysis [1].

According to previous findings, NLP and data mining are widely used techniques in sentiment analysis [2], hate speech identification, and assessing Urdu speeches [3] and movie reviews [4]. Many of these technologies are global and have a well-known influence in different aspects of human life.

Social media is one of the most popular digital platforms in recent years. These server platforms are frequently used and transmit enormous quantities of data. However, it has also been associated with the rise of hate speech. For instance, when a new film is launched, individuals make a lot of posts on social media about it, whether good or bad. Stop hate speech has received a lot of attention from scholars due to the increase in hate speech incidents. A notable paper [5] clarified the pivotal role of NLP in hate speech detection, showcasing how automated processes can effectively identify and flag instances of hatred within social media content.

Understanding and processing human-constructed natural language strictly demand NLP, which is fundamental tools. First and foremost, the information produced by users on social media and review platforms contains useful knowledge for various sectors such as healthcare, industry, science, and

policy-making. The information contained in this content entails high textual content and is potential in providing information and decision-making in diverse areas. Leveraging opinion extraction algorithms and sentiment analysis techniques [6], researchers can extract valuable insights from this data bank. Notably, these algorithms play a crucial role in enhancing the performance of text classification models during the feature extraction phase [7]–[9]. This approach helps stakeholders understand consumer sentiment, market trends, and public opinion, providing valuable insights.

In their work, the team of scholars described in detail the current shortcomings and loopholes in the field of hate speech [10] detection. To that end, they presented a solution that aimed at overcoming these challenges, focusing on the lack of ample data needed to train accurate hate speech detection algorithms. This means that the lack of adequate labeled datasets is a major challenge in accurately detecting hate speech occurrences in social media. To overcome this challenge, the researchers encouraged the use of pre-trained models like BERT since it is effective in contextual text analysis [11]. Their approach is quite sensitive to changes in data over time and is exemplified by the ALBERTo model trained on the Italian hate speech dataset. However, by using only the posterior half of the data for training the models, their accuracy significantly increases. The study further establishes the differences between hate speech and offensive language, as well as, between them. Their method improves text classification through using users' entries on social media platforms and metaheuristic algorithms; therefore, their method can be considered a promising approach to counter hate speech on social media platforms.

Transfer learning is a highly effective technique which utilizes information from the unlabeled data to enhance the overall performance on tasks that have a small number of labeled samples. It shows that prior information makes it possible to obtain high accuracy even given a small number of labeled data points. This is especially the case in NLP, where transformers outcompete others such as traditional ML and DL techniques. Authors specify that transfer learning obtains knowledge from related domains so that ML systems adapt to new tasks with higher accuracy. Thus, it has a minimized demand for active human intervention and performs better than active and supervised learning. For example, a system trained to recognize apples can be tweaked to recognize pears cutting down on the data needed and time taken to train the system.

Social media platforms wield significant influence and provide users with a platform to freely express their thoughts and opinions. Leveraging this platform, researchers have used advanced machine learning models to distinguish between real and fabricated information, especially in relation to news about COVID-19. Among these models, transformers have proven to be particularly effective tools, providing superior performance in distinguishing between real and fake content.

News dissemination is of enormous importance and serves as a crucial source of information that keeps people informed and engaged. However, the spread of fake news dates back to the 18th century [12], and the advent of the Internet has exacerbated its spread, primarily through the widespread use of social media channels. Detecting the authenticity of news articles poses a huge challenge, and existing approaches show limitations. Researchers are developing hybrid methods to improve the accuracy and consistency of fake news detection and overcome current limitations.

In the recent context, research has diligently used various optimization techniques to improve model performance, effectively reduce losses, and improve overall accuracy. Using state-of-the-art algorithms, the research has refined the model parameters to achieve optimal results, ensuring the robustness and reliability of the predictive results. Additionally, to enhance efforts, the study used ChatGPT as a valuable resource during our experiments and papers. Their assistance was essential to refine methodologies, interpret results, and effectively articulate findings.

The remainder of the document is organized as follows: it reviews the relevant literature, explains different classification algorithms, describes the methodology used, discusses the experimental results, and presents the conclusions and future directions.

II. Relevant Literature

In modern research, transfer learning stands as a fundamental technique that struggles for greater precision in various domains. Researcher explore different transformer iterations to advance their investigations effectively. In their recent study, the researchers delved into the latest advances in TL within the field of NLP [13]. Their comprehensive analysis involved examining new ML and DL methodologies, as well as exploring modern TL approaches.

The researchers observed that TL techniques have introduced innovative dimensions in various NLP tasks, particularly benefiting scenarios with limited training data. By leveraging pre-trained models and subsequent fine-tuning, TL models demonstrates superior performance over state-of-the-art NLP methods. Pre-trained and fine-tuned models act as a powerful transfer mechanism, but their efficiency can vary from task to task, prompting the development of new techniques to address complex challenges [14].

As an alternative, the researchers have introduced an adapter module with pass-through. These modules facilitate the creation of scalable models, where each task addition requires minimum trainable parameters, avoiding the need to revisit previous tasks. By maintaining a high degree of parameter exchange, the original model parameters remain fixed, increasing stability. Specifically, the evaluation of BERT on 26 different classification tasks, compared to GLUE. By including on 3.6% additional parameters per task, GLUE demonstrated excellent performance by outperforming traditional methods that refine all parameters indiscriminately.

Recent research shows the effects of misinformation about COVID-19 [15]. It highlights how social media contained many unreliable messages that caused widespread panic. The

research present a strong way to detect fake news, called CT-BERT (COVID-Twitter-BERT) which achieved an impressive F1-score of 98.69.

Researchers also developed a method to check whether social media posts about COVID-19 were credible [16]. Their approach used TL models such as BERT, ALBERT and XL-NET. When tested on a large dataset, it received a remarkable F1 score of 98.55 and ranked 5th out of 160 teams. By carefully dividing the dataset, the study create a solid way to test things. Highlights the importance of disseminating reliable information during crises.

Social media makes finding news very easy nowadays, but unfortunately, fake news also spreads like wildfire. To combat this, experts studied things like the actual content, who shared it and their profiles, using these clues to detect fakes [17]. But this approach requires reliable datasets, which are difficult to obtain. Therefore, a new unsupervised method called Unsupervised Fake News Detection Technique (UFD) was developed as a solution. UFD ensures that users can trust real news and not fake messages. It was tested using 12,800 short statements and 1,627 American election articles. UFD outperforms other methods and offers the best in precision, recall, F1-score and accuracy. Accuracy rates were high: 0.759 on LIAR and 0.679 on Buss Feed data.

In an analysis evaluating strategies for recognizing fake news during the COVID-19 pandemic, researchers have explored deep study and change control strategies [18]. Analyzing a dataset of social posts and articles with 10,700 entries, they engaged several algorithms, along with Bi-LSTM Attention, Hierarchical Attention Network (HAN), Distil-BERT and BERT-base. Their goal is to distinguish between real and fake news. By focusing exclusively on news content, they ignored ability signals form personal development or social media, which are not always helpful. Interestingly, pre-training BERT and Distil-BERT on COVID-19 tweets provided superior overall performance compared to models that were first-class fitted only on the dataset. Especially, the version trained manually with the BERT case performed better than others carefully trained using the COVID-Twitter-BERT version.

In reference [19], researchers delved into the prevalent effect of social media in our daily lives, shedding light on its potential dangers. They addressed the difficulty of misinformation head-on, especially when it comes to COVID-19 news. Their modern approach to distinguishing real information from fake earned them much praise, and their version earned a commendable F1-score and secured second place in the rankings.

Using a large number of datasets form platforms such as Facebook, Twitter and Instagram, they carefully divided them into teaching, testing and validation units. Through rigorous experimentation with various base and transformer models, researchers validated the superiority of their RoBERTa and Electra models, achieving extraordinary F1 ranks of 0.9864 and 0.9827 respectively on the legitimate test set.

In this groundbreaking study, researchers proposed a new approach to political text classification that leverages the insights of human experts through hard annotation of political manifestos [20]. This annotated corpus serves as invaluable training data for an advanced classifier, which is then applied to the dynamic landscape of COVID-19 press conferences. By ingeniously combining convolutional neural networks (CNN) with BERT transformer technology, researchers achieve remarkable results.

Among the different models examined, the fusion of CNN and BERT produced the highest accuracy. Through rigorous experimentation, named M1, M2, M3 and M4, it is revealed that M4, which embodies the synergy of CNN and BERT, emerges as the most effective, with superior accuracy and F1-socres.

In an innovative investigation on sentiment evaluation inside finance, a committed staff of researchers tested latest improvements within the subject [21]. Their thorough research encompassed diverse methodologies, from traditional lexicon-based totally strategies to modern-day transformer fashions. Among those, transformers emerged as the undisputed champions, boasting unheard of accuracy across the board.

The group scrutinized an array of transformer models, along with BERT, FinBERT, and XLNet, amongst others. Remarkably, Facebook's BART stood out with an awesome F1 score of zero.85, showcasing its prowess in decoding complex financial sentiments. This have a look at indicates a sizeable leap ahead in know-how and harnessing the strength of herbal language processing in financial analysis.

BERT, while trained on vast corpora like BookCorpus and Wikipedia, grants attractive effects in many natural language processing regions [22]. However, its efficacy hesitates in certain domains and responsibilities, revealing room for enhancement. Addressing this hole, researchers brought BERT4TC, a sophisticated new release tailored for textual content type. This upgraded version boasts enriched area-unique and task-orientated expertise.

Rigorously evaluated on publicly handy datasets, BERT4TC shines. Outperforming conventional feature-based procedures and even some best-tuned techniques, it establishes new benchmarks in multiclass classification. Through meticulous layout and thorough evaluation, BERT4TC heralds a fabulous step closer to advancing NLP talents.

In today's society, the internet has been seamlessly integrated into our lives and has become an essential part of our routines [23]. While it serves as a trusted source of knowledge and information, it has highlighted some concerning issues. The prevalence of harmful language on the internet underscores the importance of taking action. Advances have been made in the field of NLP through the use of ML techniques. An example is the DeepHate model, which was developed to efficiently analyze data. By training with datasets, you can successfully detect patterns of hate speech despite the challenges. With transfer learning as its basis, the model does not determine the meaning of words. It also understands sentence structures. Studying datasets such as English tweets (consisting of 37,520 tweets) represents a step forward in the

fight against online harassment.

III. Material and Methods

The overall methodology used to classify COVID-19 news as real or fake is demonstrated in Fig. 1. It includes dataset collection, data preprocessing, data splitting and model evaluation.

A. Dataset

The recent research study analyzed the "COVID-19 English tweets" dataset, crafted by [55]. The dataset was properly labeled with 3360 real tweets, 3060 fake tweets, and a total of 6420 tweets, it offers valuable insights into pandemic-related social media discourse.

The dataset is publicly available and contains two columns. The first column is "Tweets" which contains the textual content, while the second column is "Labels" which differentiates authentic tweets from bogus tweets. Examples are presented in Table I.

- Tweets: This column contains the original textual data representing specific tweets related to the COVID-19 pandemic. Each row contains the textual content of a particular tweet, ranging from updates, statistics, and precautions to opinions concerning the pandemic. This column contains the original textual data representing specific tweets related to the COVID-19 pandemic. Each row contains the textual content of a particular tweet, ranging from updates, statistics, and precautions to opinions concerning the pandemic.
- Labels: This column serves as a classifier by specifying the authenticity of the information contained in each tweet about COVID-19. Tweets are categorized as "real" when they contain accurate information and are labeled as "fake" when they potentially contain ambiguous information related to the pandemic.

After the collection and organization of the dataset, the next step was data preprocessing. In this step, the study primarily focused on data cleaning, which involved removing URLs, special tags, and numbers and replacing multiple spaces with a single space. Additionally, all characters were converted to lowercase to ensure consistency. These procedures ensured that the dataset was clean and ready for further processing.

Fig. 2 summarizes the contents of dataset collection.

In order to process the data, real labels were replaced with 1 and fake labels with 0, using a Python script. Fig. 3 illustrates the frequency of the top 30 most commonly occurring words.

Fig. 4 shows a word cloud generated from raw and unprocessed data. It visually represents the frequency of words in the dataset. However, further cleaning and processing may be needed for more accurate analysis and interpretation.

B. Data Preprocessing

After the collection and arrangement of the dataset, the next step to perform is data preprocessing. In this step, the study generally concentrated on cleaning of data, which primarily involved removing URLs, special tags, and numbers, and replaced multiple spaces with a single one. Additionally, in this phase all characters are converted into lowercase for consistency. This ensures the dataset is clean and ready for analysis. Fig. 5 and Fig. 6 demonstrating the frequency of most frequent top 30 words and word cloud of cleaned and preprocessed dataset. In the recent context, the research did not utilized any approach to balance the dataset, as the dataset has enough samples of both classes to perform the experimentations.

C. Training and Validation Splits

After cleaning and preprocessing, the study divided the dataset into two parts, a training set and a validation set. The training set consists of approximately 70% of the data, while the validation set contains 10% of the data. Finally the test set contains the remaining 20% of the data. Therefore, models are trained on 70% of the data, validated on 10% of the data and tested on 20% of the data. This 70:10:20 ratio ensures that the majority of the data is used to train and validate the model, with some reserved for testing its performance.

D. Model Evaluation

After completing the training, the study evaluated the model's performance using various parameters such as precision, recall, accuracy and F1-score. When evaluating classification models, the study rely on tools such as confusion matrices and heat maps. These tools help us to understand how well the nominated model is performing by comparing the predicted values with actual values.

The confusion matrix, a simple $M \times N$ matrix, contain values such as true positives, true negatives, false positives and false negatives [24]. It's like a map that guide us thorough the accuracy of our predictions and shows where the model excels and where it needs improvement.

E. Classification Algorithms

This study explored seven classification models: BERT-base, XLM-RoBERTa-base, ALBERT-base, and DistilBERT for TL, along with three LLMs such as - GPT2, T-5, and XLNet. LLMs are advanced machine learning models designed to understand, generate, and manipulate human language. They are typically built using deep learning techniques and trained on vast amounts of textual data.

In the study, fine-tuned LLMs were used. The parameters of these selected TL and LLM are initialized with their default settings except some of them are altered as per need, such as epochs, learning rate and batch size of the models. For example, the default learning rate for GPT-2 model is 5×10^{-5} but in the current research context the learning rate is initialized with 2×10^{-5}.

First, the TL models were evaluated on the dataset to assess their performance. Optimization techniques were then applied to further improve their results. By comparing model effectiveness before and after optimization, the study aimed to improve performance and achieve better results.

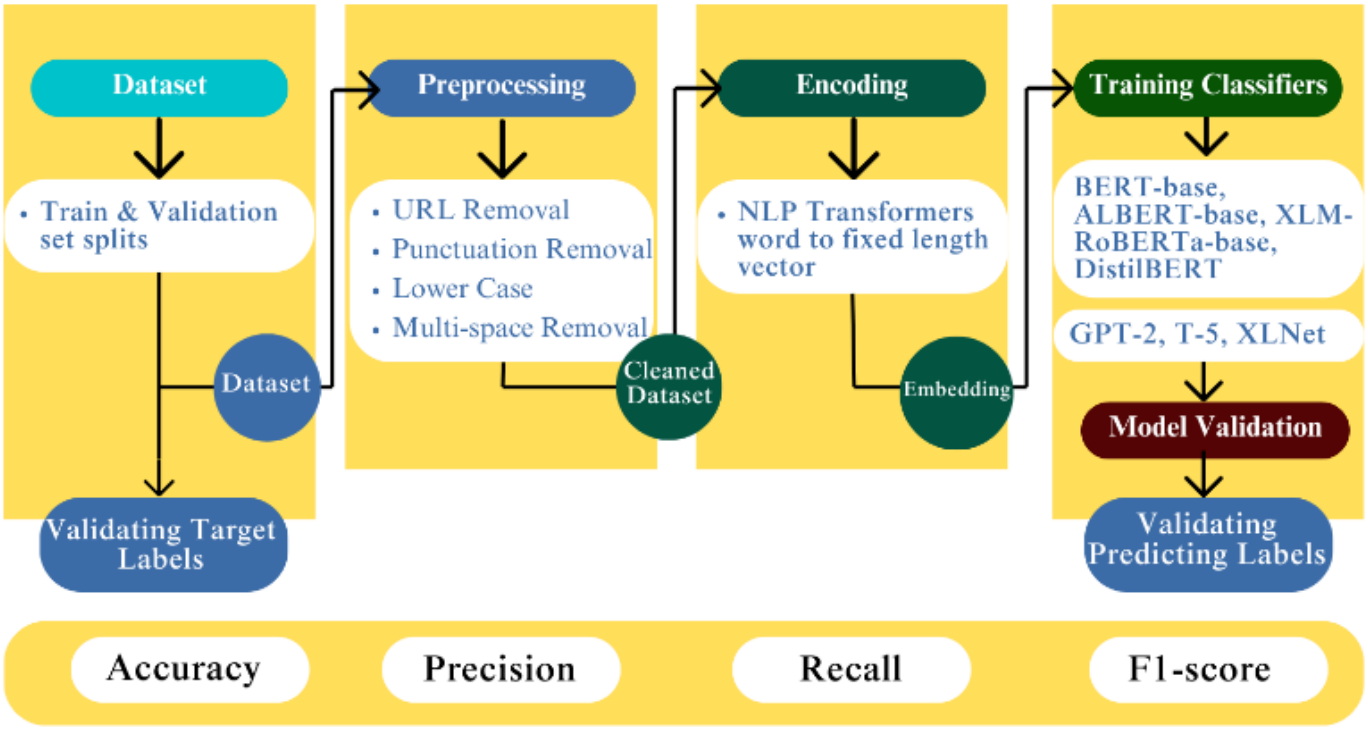


Fig. 1. Block diagram

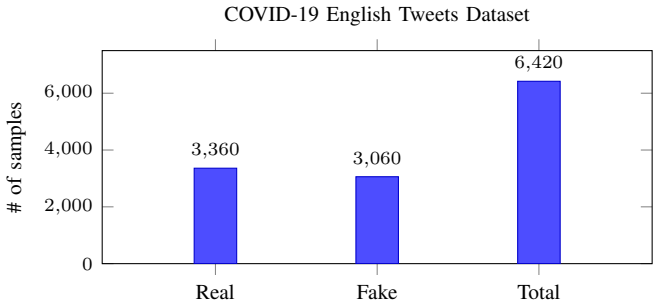


Fig. 2. Structure of dataset

1) BERT-Base: BERT, introduced by researchers in [25] is an abbreviation for Bidirectional Encoder Representations of Transformers. The main goal of BERT is to extract meaning from data that does not have labels. BERT intelligently takes in to account appropriate contexts at all levels, making it a powerful tool in natural language processing.

Despite its simplicity, BERT delivers results for machine learning tasks. Adapting a BERT model for a task is simple; just add a layer. This flexibility allows BERT to effortlessly address challenges, demonstrating its versatility and intelligence effectiveness. BERT uses a Masked Language Model (MLM), words are hidden in the input text and the model predicts them based on the surrounding words. By considering both left-wing and right-wing contexts, bidirectional understanding is trained.

Additionally, it combines MLM with Next Sentence Prediction (NSP) for better understanding. BERT-Base requires less processing and calculation time. In certain tasks like spotting fake news, refining content and distilling knowledge, the approach shines. But it is not suited for fuzzy data or text mining [24]. In literature [26], BERT is used to detect fake news. The reference [27] explored it for content improvement and the model shows its encouraging results in the particular field. Subsequently, BERT, as seen in studies [28], [29], also excels in sentiment analysis, proving its worth across diverse applications.

2) ALBERT-Base: In 2019, Toyota Technologies and Google Research partnered to uncover ALBERT [30], a compact successor to BERT. By trimming parameters, ALBERT accelerates BERT's training speed and reduces memory usage. ALBERT perform superior in various classification tasks and excels in sentiment analysis, despite its nominal parameter count [24].

In a study by [31], ALBERT transformer method was employed to detect sarcasm on Twitter. The approach was also tested for fake news detection, outperforming other model as noted in [32]. Additionally, ALBERT was evaluated for COVID-19 question answering, achieving a remarkable match score of 13.04 in [33]. Furthermore, [34] utilized ALBERT for analyzing medication recommendations shared on social media platforms.

3) DistilBERT: In 2019, [35] disclosed DistilBERT, a sleeker, swifter, and more economical variant of BERT. 60% faster speed, with a 40% reduction in size and 97% language comprehension, researchers swiftly embraced its efficiency

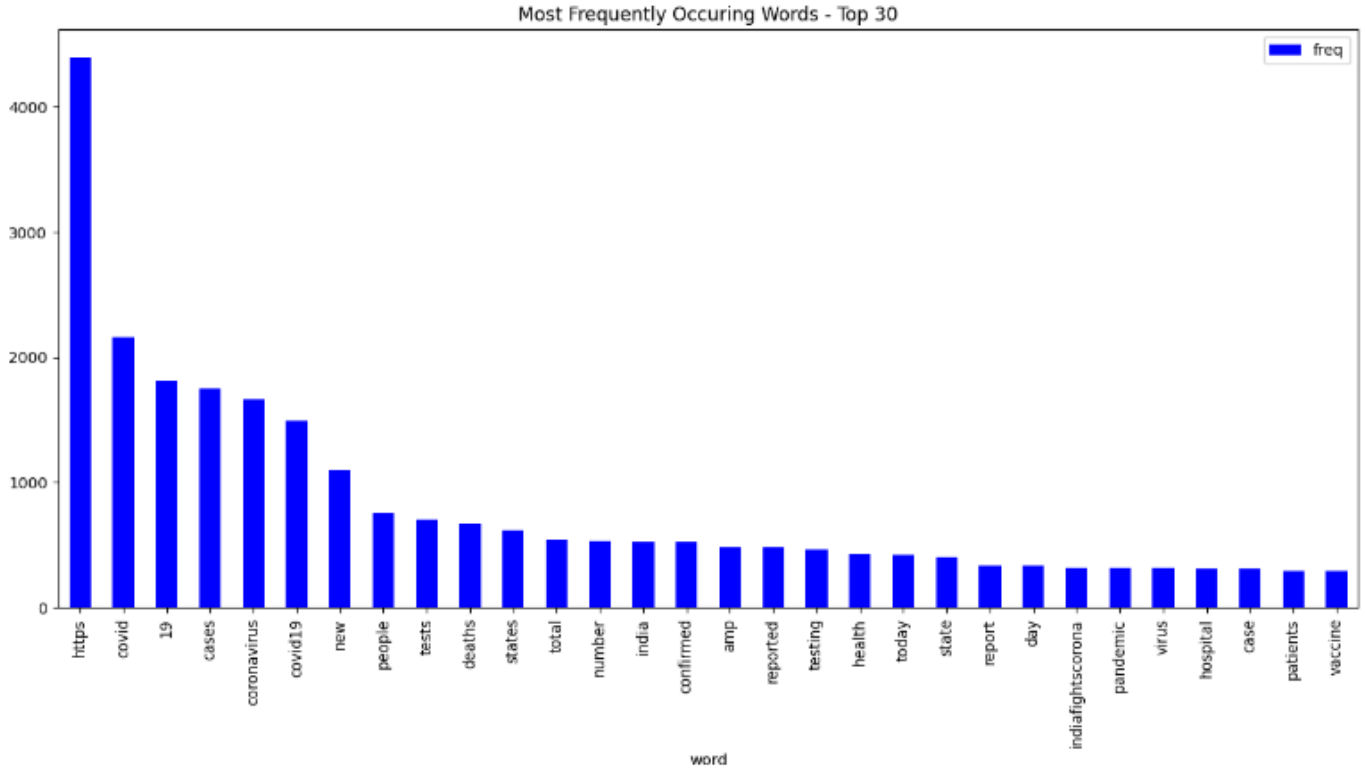


Fig. 3. Frequency of top 30 frequent words of raw data

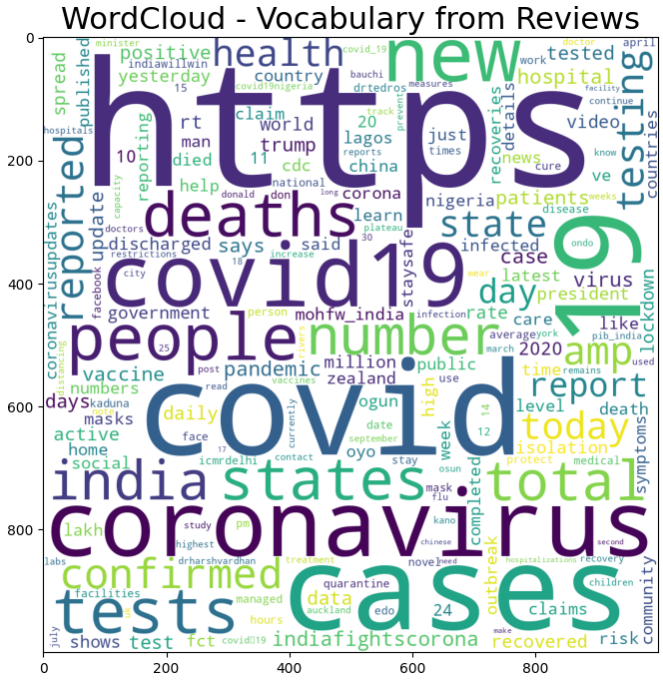


Fig. 4. Word cloud of raw data

[24]. In their study, the authors [36] examined the ability of DistilBERT, a simplified version of BERT, to classify social and political contexts. The findings showed promising results, demonstrating the effectiveness of DistilBERT in this area.

Another study [37] found dramatic improvements in performance by adding linguistic comprehension and incorporating transfer learning models. In addition, the ensemble model significantly increased accuracy of the existing model. Additionally, the researchers [38] used current variant of BERT to identify health information and name the institutional identification task, which yielded a promising 50% increase in search accuracy. In a study, researchers [39] enhanced DistilBERT for answer selection, improving performance by 0.6%.

Another team [40] retrained DistilBERT on universal dependencies to create a voice shopping assistant. Both approaches show promising advancements in natural language processing.

4) XLM-RoBERTa-Base: In the literature [41], the authors introduced the recent model by training it across a hundred languages using two terabytes of filtered data. Their collaboration with XLM-R resulted in a decent 23% improvement over various transformer models. In study [46], researchers explored this approach for multi-language sentiment analysis, reporting a successful outcome with the model achieving a worthy F1 score.

In their study for EVALITA 2020, researchers in [47] utilized neural networks for classification tasks, achieving an impressive F1 score of 0.798 on a hate speech dataset. Meanwhile, [48] demonstrated the superior performance of XLM-RoBERTa in word context disambiguation, surpassing

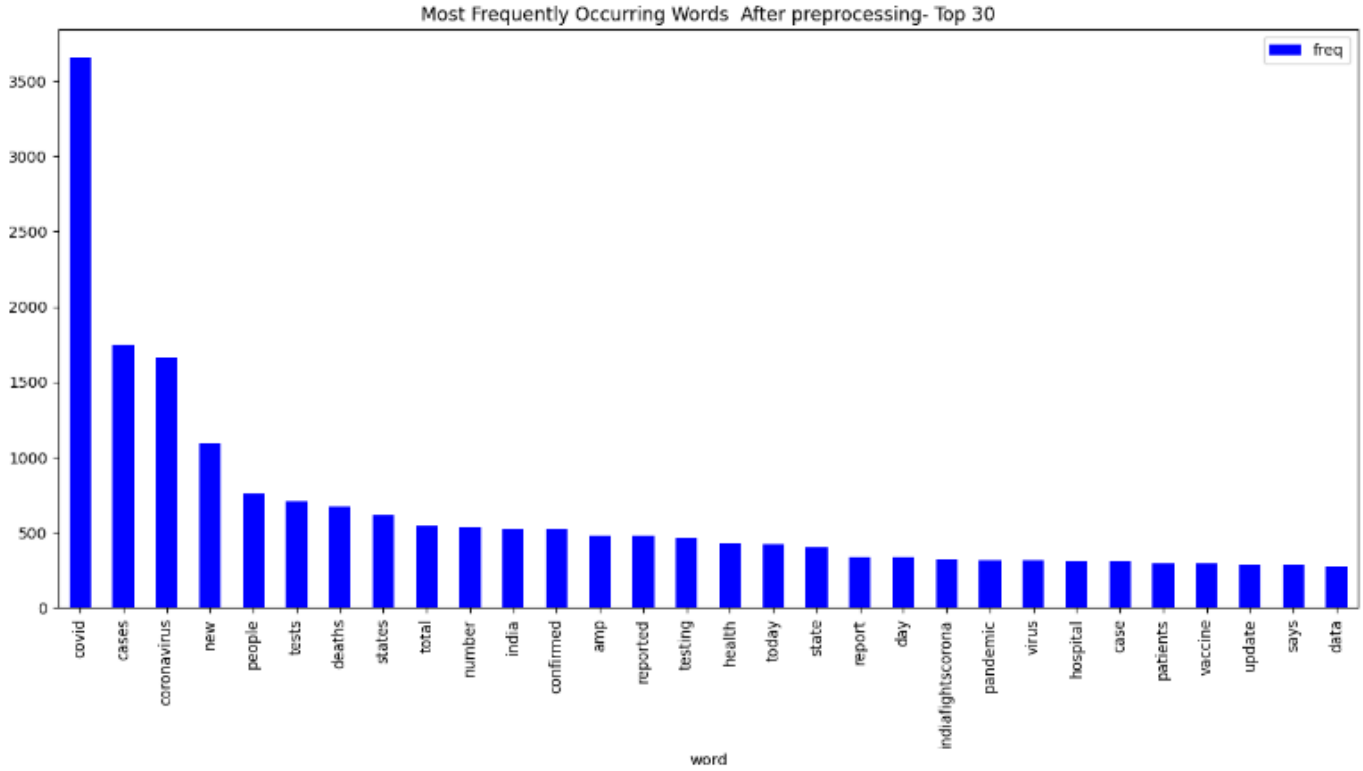


Fig. 5. frequency of top 30 frequent words of processed data

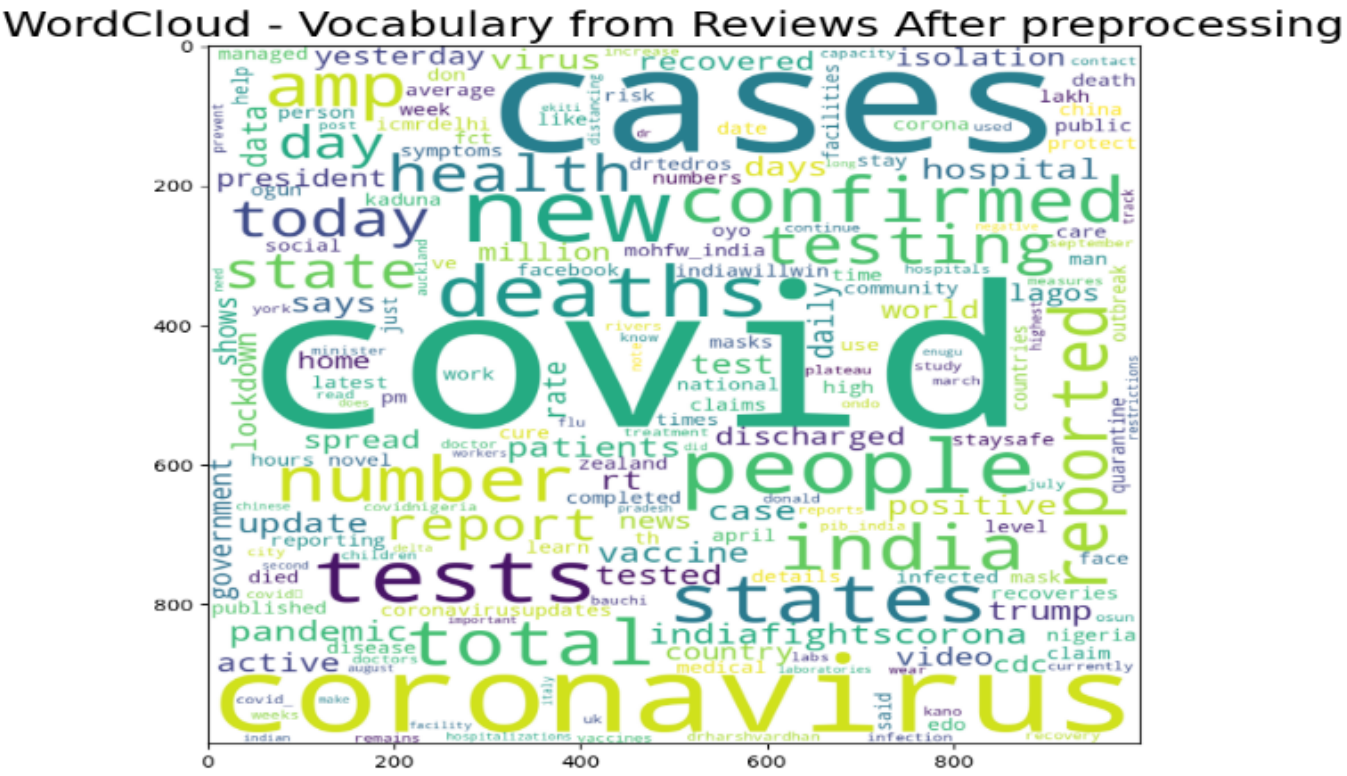


Fig. 6. Word cloud of processed data

other tested methods.

The researchers of [45] combined XLM-RoBERTa with DPCNN to spot offensive language, yielding impressive outcomes. They also applied this strategy for hope speech detection, showcasing encouraging results with F1 scores of 0.92, 0.84, and 0.59 for English, Malayalam and Tamil languages, respectively.

5) GPT-2: The GPT-2 (Generative Pre-trained Transformer 2), is like a creative writer trained to generate text. With its autoregressive ability, it crafts sentences word by word, capturing context and creating flowing narratives. It is a powerful tool for producing various and clear text across several topics [49].

6) XLNet: It is a state-of-the-art language model by Google AI and Carnegie Mellon University, pushes the boundaries of language understanding. XLNet improves on BERT and GPT by using Transformer-XL for its core architecture. Unlike BERT, which masks input tokens and doesn't fully capture bidirectional contexts, XLNet learns bidirectional contexts more effectively by maximizing expected likelihood.

XLNet uses an autoregressive approach, integrating Transformer-XL into its pre-training process to address the limitations of BERT and enhance overall performance. This method overcomes the issues of BERT with masked positions and better models dependencies in the data [58].

7) T-5: The T5 model, created by Google AI, stands out in the world of language processing. Text-to-Text transfer transformers are becoming popular because of their strong performance and simplicity in handling various tasks by treating them as text-to-text problems.

T5 uses an encoder-decoder transformer, is pre-trained through an unsupervised span corruption task. While T5 has been effectively used for many natural language processing tasks, the methods for extracting high-quality text representations from T5 are still not well understood [59].

F. Optimization Techniques and Fusion

In recent research, various optimization techniques are explored to boost model performance and achieve optimal outcomes. According to the [50] fusion approach, researchers focus on merit-based late fusion, where they combine classification score from different models using various weight selection and optimization methods.

A separate validation set is used to calculate the classification error of the fitness function for each method. The weights are selected based on the previous success of these methods in similar tasks, thereby supporting the reliability and effectiveness of the optimization approach [51].

1) Weighted Fusion: It improves prediction accuracy by giving more weight to better performing models. Assign higher weights to models with greater precision or reliability, so that the final result reflects the strength of each model. By averaging predictions with these weights, this method balances the influence of each model, resulting in a more reliable and accurate overall result.

2) PSO Fusion: Particle Swarm Optimization (PSO) fusion [52], is a widely used optimization technique known for its effectiveness and versatility in various tasks. It has been applied for hyper-parameter tuning in Neural Networks [54] and Federated Learning (FL) [53]. PSO starts with a group of diverse solutions and improves them iteratively based on a fitness function. It updates particle positions and velocities using key parameters like personal and global best. It remains effective for many applications despite these challenges [50].

3) Nelder-Mead Algorithm: The Nelder-Mead algorithm, introduced in 1965 by Nelder and Mead, is one of the oldest and widely used optimization method. It is effective and popular in various fields, with support in numerous statistical and machine learning libraries, similar to PSO [51]. It is ideal for n-dimensional problems and excels at finding the maximum/minimum of an objective function, such as minimizing classification errors to obtain better weighted results based on validation data [50].

4) Powell's Method: Powell's method, a popular search and optimization approach, relies on the censored maximum gradients technique. Its simplicity and flexibility make it applicable across various domains. Like PSO, it iteratively refines an initial presumption, moving towards function minima.

While processing, the method hunts and computes the distance in a worthy track. Despite its effectiveness, Powell's method has limitations, such as assumptions about step size. Overall, it offers a straightforward yet effective strategy for problem-solving and optimization [50].

5) BFGS: The BFGS algorithm, known as Broyden-Fletcher-Goldfarb-Shanno, is like an experienced guide to finding the best route in a complex landscape. It is an iterative method that helps minimize or maximize functions in multidimensional spaces, making it ideal for complicated optimization tasks. This method is useful when the gradient of the function is known and excels at smooth, well-behaved problems even when the function is difficult to decipher. It is widely adopted and is a tool of choice to address unlimited optimization challenges.

6) Trust Region Constrained (trust-constr) Algorithm: The Trust Region Constrained algorithm minimizes or maximizes a function while respecting constraints. It is great for problems with both equality and inequality constraints, using trust regions for efficient optimization. This method is very efficient for tackling difficult optimization tasks, especially when dealing with rules and complex functions that behave predictably.

IV. Experiment and Results

In this section, the study will elaborate the results of each investigation performed on available dataset by means of all performance parameters as well as the heat maps.

Table II displays the outcomes of four TL classifiers without applying the optimization techniques. While, Table III presenting the grades of utilized optimization techniques. Table IV illustrating the results of three LLMs when they are investigated on the prescribed dataset. Table V displays the

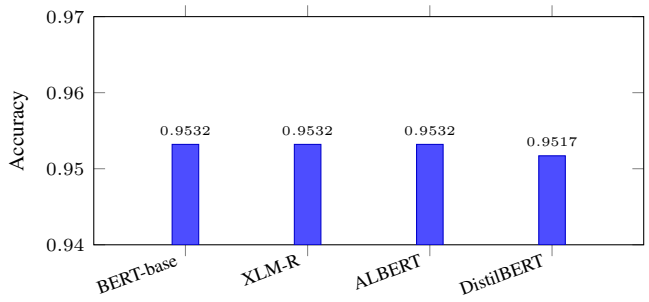


Fig. 7. Accuracies of transfer learning classifiers

comparative analysis of the projected work with state-of-the-art approaches.

Fig. 7 is displaying the accuracies of all TL models applied on COVID-19 English Tweets dataset. In Fig. 8, the values of performance parameters of each TL model are presented. Fig. 9 is demonstrating the results of applied optimization techniques. Fig. 10 shows the accuracies of LLMs which are applied on same dataset. Fig. 11 showcasing the values of all parameter values of each LLM.

Figs. 12–15 demonstrating the heat maps for TL models applied for experimentations whereas, Figs. 16–21 are illustrating the heat maps of all the applies fusion/ optimization techniques in the proposed work.

Figs. 22–24 are visually explaining the heat maps for LLMs, utilized in the study.

V. Discussion

Transformer-based models demonstrated remarkable performance on the COVID-19 English Tweets dataset. GPT-2 was identified as the best-performing model, achieving an accuracy of 96.31%. It also achieved precision, recall, and F1-score values of 96.12%, 96.10%, and 96.00%, respectively.

XLM-RoBERTa-base and ALBERT-base followed closely, with each model achieving an accuracy of 95.32%. The results of all experiments conducted in this study clearly showed that the transformer-based models and LLMs performed better than the state-of-the-art machine learning models when considering the performance metrics and time complexity.

When several models produce similar results, this may be attributed to the use of a small and balanced dataset.

Additionally, the use of similar hyperparameters or initialization methods may cause the models to converge toward similar solutions, resulting in nearly identical performance.

VI. Conclusion

In this research study, four transfer-learning (TL) models, namely BERT-base, XLM-RoBERTa-base, DistilBERT, and ALBERT-base, and three large language models (LLMs), namely GPT-2, T5, and XLNet, were applied to the COVID 19 English Tweets dataset for binary text classification. The dataset was obtained from a reliable repository and contains real and fake English-language tweets.

All TL models and LLMs were evaluated using performance metrics such as accuracy, recall, precision, and F1-score. Several optimization techniques were also explored, including weighted fusion, PSO fusion, Nelder–Mead fusion, Powell's method, BFGS fusion, and trust-region constrained fusion.

Based on the results of this research, GPT-2 was identified as the best-performing model. It achieved an accuracy of 0.9631, a precision of 0.9612, a recall of 0.9610, and an F1-score of 0.9600. These results further demonstrate its effectiveness in accurately classifying real and fake news.

In future work, we plan to conduct experiments using larger datasets for multiclass classification. We will also explore datasets in different languages to improve text-classification performance. Additionally, we plan to use Twitter's live stream to collect real-time tweets for sentiment analysis and to investigate data from other social media platforms.

VII. Author Contributions

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TABLE I
TWEETS OF COVID-19 ENGLISH TWEETS DATASET

Tweets	Labels
The NBA is poised to restart this month. In March we reported on how the Utah Jazz got 58 coronavirus tests in a matter of hours at a time when U.S. testing was sluggish. https://t.co/I8YjjrNoTh https://t.co/o0Nk6gpyos	Fake
We just announced that the first participants in each age cohort have been dosed in the Phase 2 study of our mRNA vaccine (mRNA-1273) against novel coronavirus. Read more: https://t.co/woPlKz1bZC #mRNA https://t.co/9VGUoJu5cS	Real

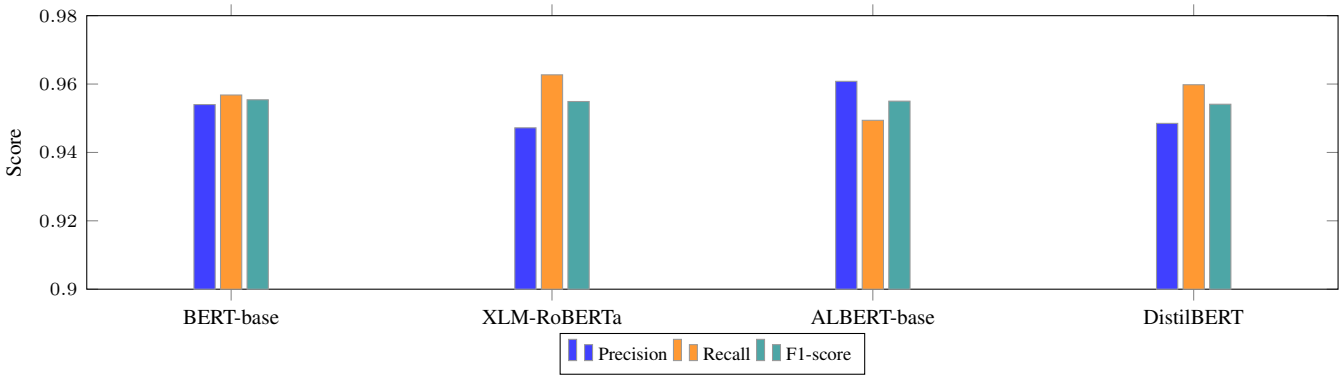


Fig. 8. Parameter values of TL models

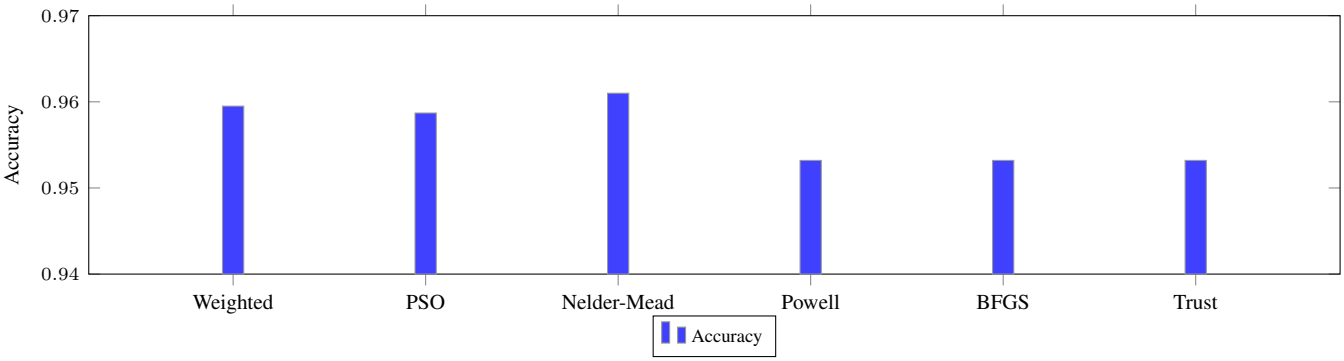


Fig. 9. Accuracies with optimization techniques

TABLE II
TRANSFER LEARNING MODEL'S RESULTS

Model	Accuracy	Precision	Recall	F1
BERT-base	0.9532	0.9540	0.9568	0.9554
XLM-RoBERTa -base	0.9532	0.9540	0.9568	0.9554
ALBERT-base	0.9532	0.9540	0.9568	0.9554
DistilBERT	0.9517	0.9485	0.9598	0.9541

TABLE III
OPTIMIZATION TECHNIQUE'S RESULTS

Method	Accuracy	Precision	Recall	F1
Weighted	0.9595	0.9599	0.9627	0.9613
PSO	0.9563	0.9723	0.9434	0.9577
Nelder Mead	0.9610	0.9627	0.9627	0.9627
Powell's method	0.9532	0.9608	0.9494	0.9550
BFGS	0.9532	0.9608	0.9494	0.9550
Trust Constraint	0.9532	0.9608	0.9494	0.9550

TABLE IV
LARGE LANGUAGE MODEL'S RESULTS

Method	Accuracy	Precision	Recall	F1
GPT-2	0.9631	0.9612	0.9610	0.9600
T-5	0.9299	0.9356	0.9314	0.9310
XLNet	0.9532	0.9540	0.9568	0.9554

TABLE V
COMPARISON OF STATE-OF-THE-ART METHODS WITH PROPOSED WORK

State-of-the-art methods		Proposed
[56]	[57]	GPT-2
0.7900	0.6100	0.9631

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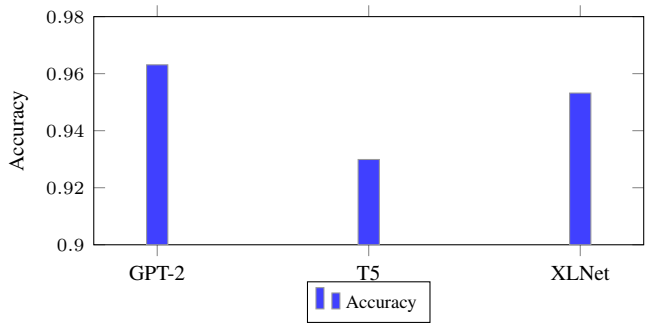


Fig. 10. Accuracies of large language models

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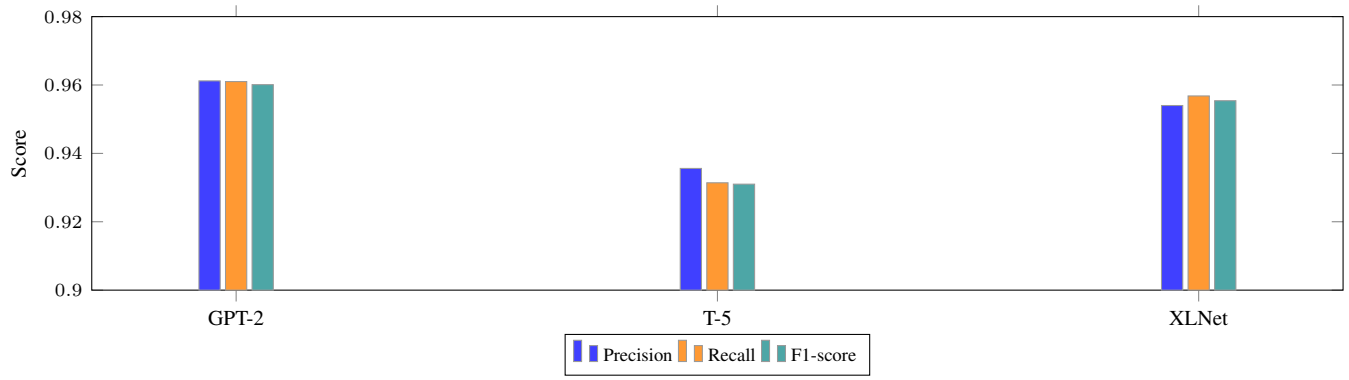


Fig. 11. Parameter values of large language models

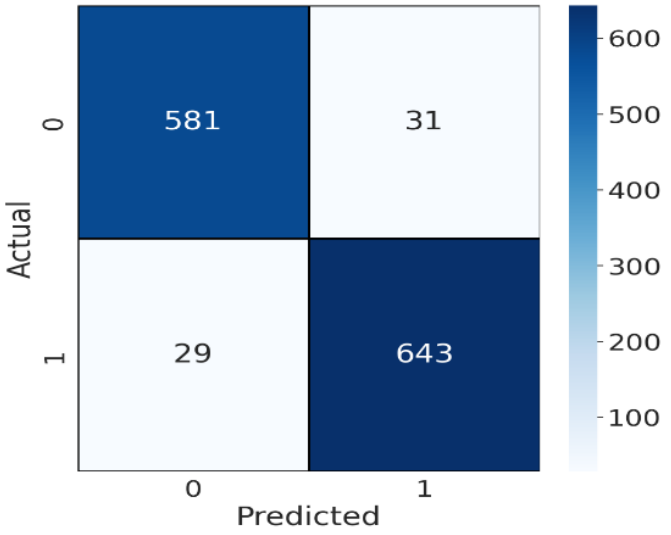


Fig. 12. BERT-base

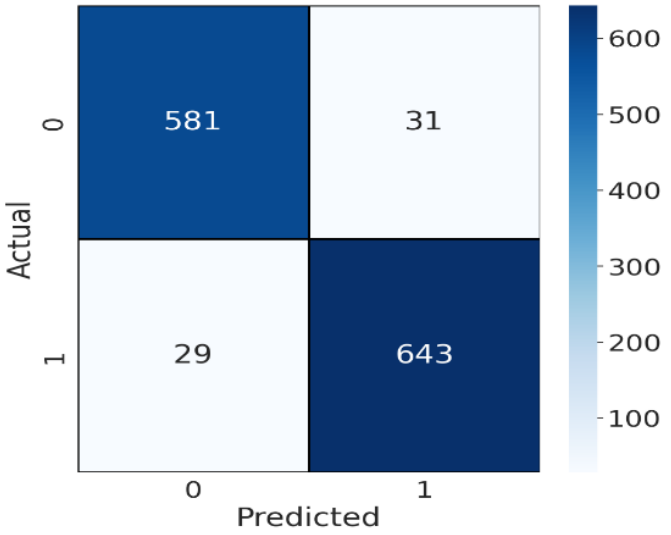


Fig. 13. XLM-RoBERTa

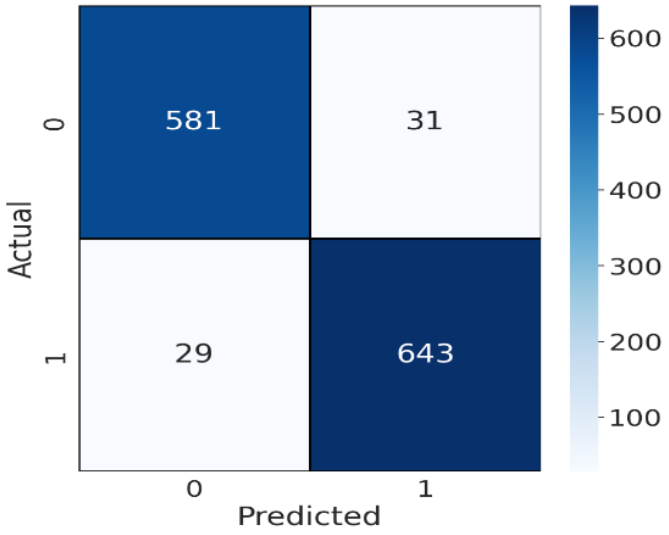


Fig. 14. ALBERT

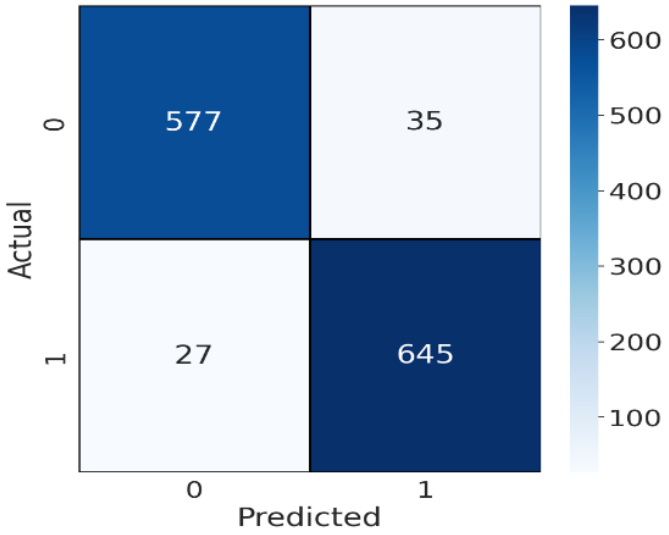


Fig. 15. DistilBERT

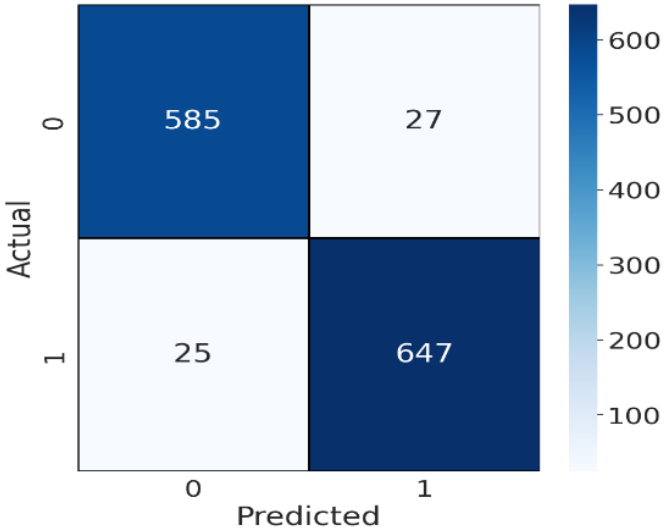


Fig. 16. Weighted fusion

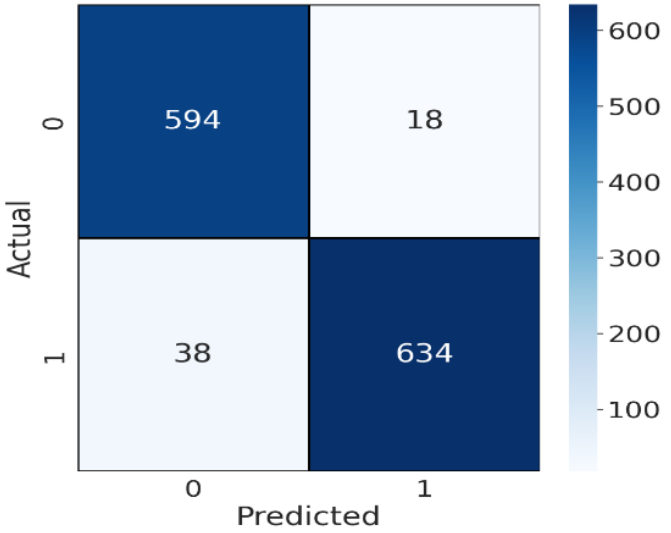


Fig. 17. PSO fusion

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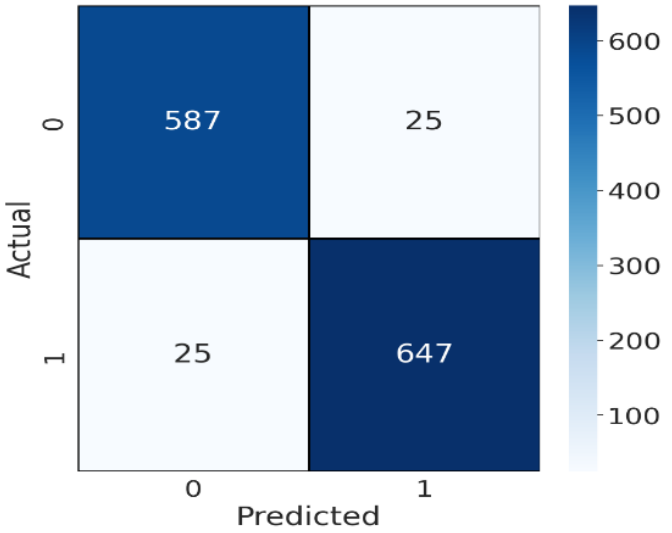


Fig. 18. Nelder-Mead fusion

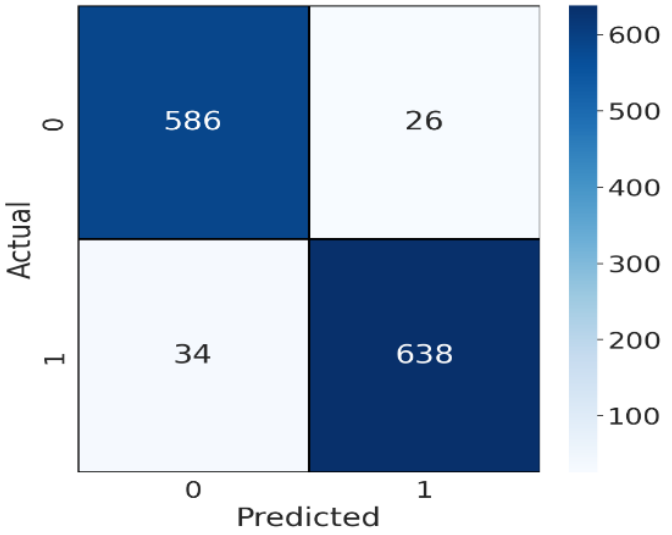


Fig. 19. Powell fusion

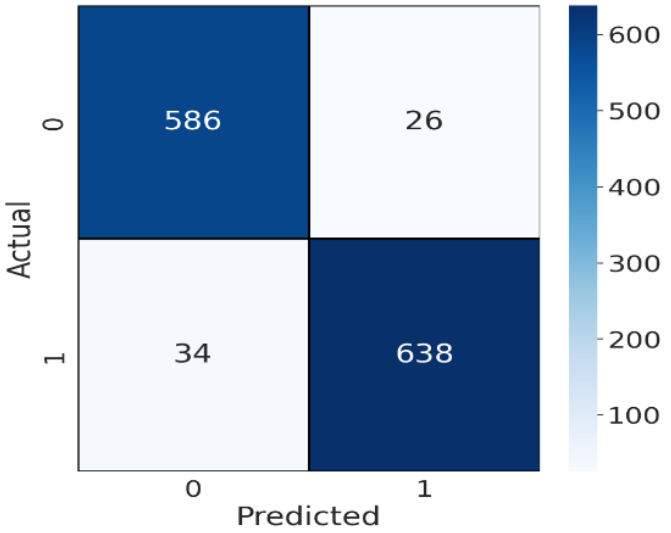


Fig. 20. BFGS fusion

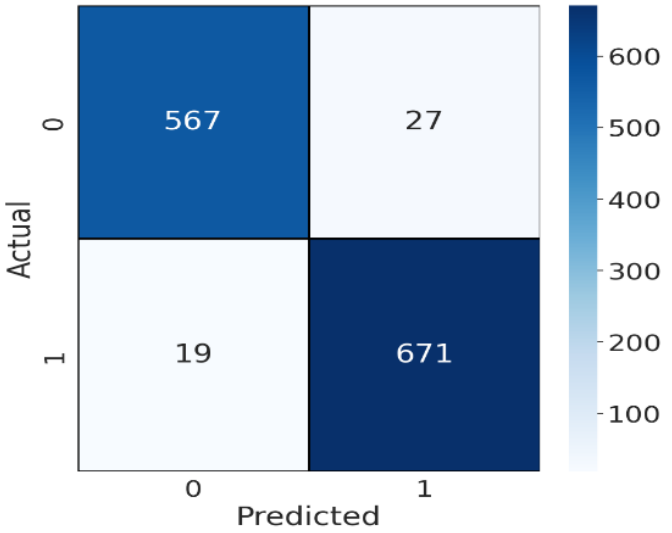


Fig. 22. GPT-2

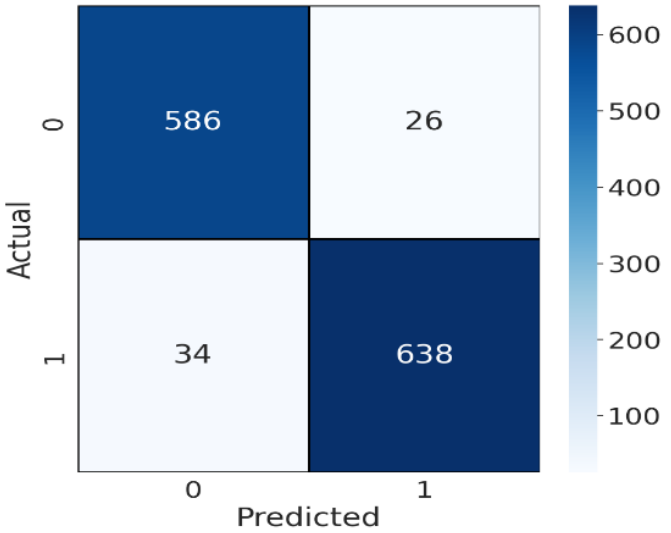


Fig. 21. Trust-Constr fusion

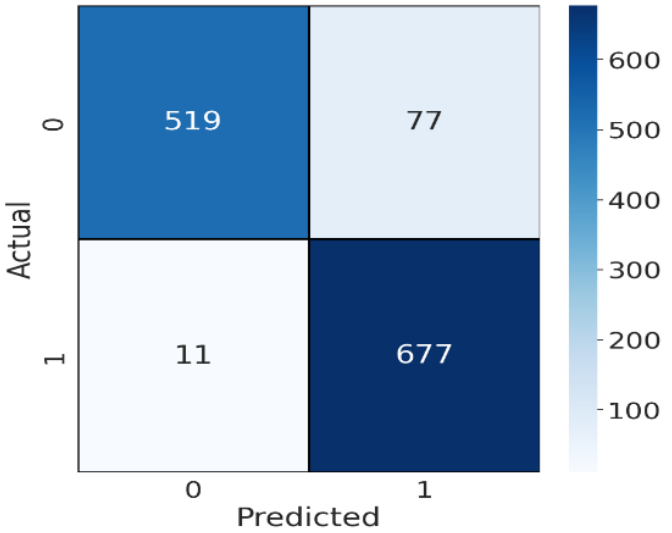


Fig. 23. T-5

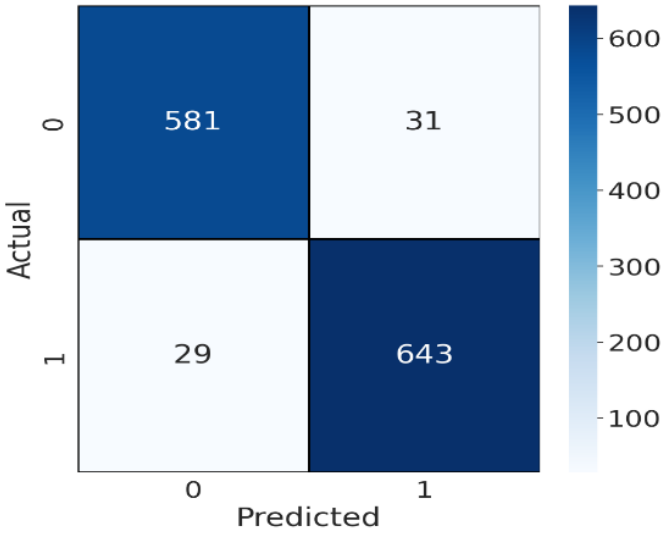


Fig. 24. XLNet